

# Video2Force: Experiencing Object Motion in Video with Dynamic Force Feedback based on Bio-Inspired Sensing and Processing

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## ABSTRACT

In this work, we propose Video2Force, a framework that can estimate the video object motion and its resulting “force” by leveraging the emerging bio-inspired vision sensors for low-complexity and low-latency visual information processing. While the user watches video, the estimated force is “delivered” to the user’s sensation with dynamic force feedback via emerging haptic gloves, synchronized with the video content. Consequently, the user is enabled to experience the “virtual existence” of the video object and its motion. Preliminary experimental results demonstrate that Video2Force is feasible and can enhance the users’ presence in video experience.

**Index Terms:** Human-centered computing—Empirical studies in HCI; Human-centered computing—Virtual reality

## 1 INTRODUCTION

Haptic feedback can increase the user’s presence in a virtual environment [1]. In this work, we propose Video2Force, a framework that can smartly estimate the object motion and its resulting “force” that appear in video, and subsequently maps the estimated force to emerging haptic devices such as “SenseGlove” [2]. By this means, the users will then be allowed to “touch” and “feel” the video object and its motion via dynamically computed force feedback, synchronized with the video content.

To make Video2Force a reality, it is essential to realize real-time and high-precision object tracking for our force estimation. However, the limited frame rates of traditional cameras are not precise enough for tracking fast-moving objects. Motivated by the principle of human visual system, the emerging bio-inspired “event” sensing technology [3] provides new possibilities to address the above issue. Instead of capturing fixed-rate frames of pixels, the bio-inspired vision sensors only measure and store the per-pixel brightness changes, which are so called “events”, in an asynchronous manner [3]. Therefore, these sensors are also named as “event cameras”. The event cameras can offer several advantages over traditional cameras including low data rate (around 1MB/s), low-power processing (around 1mW), and very high temporal resolution (up to 1Mhz). Therefore, with event cameras [4], we can track objects asynchronously in event domain with extremely low-latency, allowing to track in the “blind” time between the frames captured by a traditional camera for higher

tracing precision. The block diagram of Video2Force is shown in Figure 1 (and the technical details are described in Section 3 and Figure 2). To the best of our knowledge, our work is the first exploration in bridging video motion and haptic with dynamic force feedback via the emerging event-based sensing and processing.

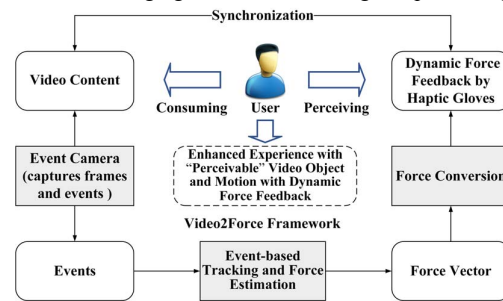


Figure 1: Block diagram of the Video2Force framework.

## 2 RELATED WORK IN BRIEF

Event-based sensing and processing have been employed in fast-moving obstacles dodging for quadrotors [5], low-latency tracking of fast-moving objects in 2D video (EKL algorithm) [4], etc. However, these methods (or similar work) cannot realize the object force estimation required in Video2Force. In the literature of haptic related research, researchers proposed some methods in improving users’ immersion with haptic feedback related to video content such as [6-7]. However, these researches are based on traditional video processing leading to high complexity and high latency, which do not satisfy our requirement of precise object tracking.

## 3 METHODOLOGY

As shown in Figure 2, the object velocity is required to estimate the resulting force of a moving object according to Momentum Theorem. In our work, the object velocity is inferred from the high-precision and low-complexity event-based object tracking. For event-based tracking, the EKL algorithm [4] employs Harris corners detectors for only global tracking but not the object-level tracking. Based on [4], our approach is developed to realize local object tracking and allow users to specify the objects of interest to be tracked. Specifically, as shown in Figure 2 (a) and its caption, we extract the object features and calculate the pixel size of the object with edge detection, binarization, etc. Then, following a maximum likelihood approach, we use the difference between the observed brightness changes from the events and the frames to estimate the motion parameters to realize event-based object tracking.

Next, as shown in Figure 2 (b) and its caption, the pixel velocity is further processed and converted to the object velocity with filtering, velocity mapping, etc. Then the estimated object velocity, together with the object mass, are used to estimate the resulting force of the object to form a “force vector”. It is to be noted that the object mass is difficult to be accurately estimated from the video

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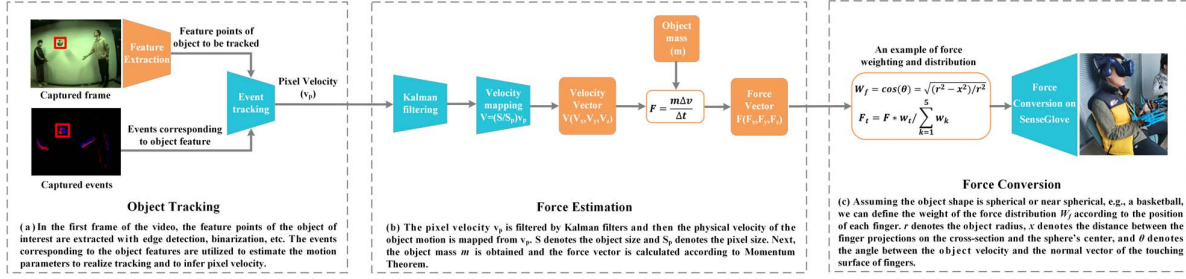


Figure 2: Diagram illustrating technical details.

directly with computer vision-based approaches. Thus, in our experiment (Section 4), we weighted the object mass with a scale for simplicity. However, there is a trend that some AI-based e-commerce companies, e.g., thetake.ai, maintain a database containing the information of million-scale products (objects) that appear in TV programs and films, which provides a possible means to obtain the mass of the video object online.

For force conversion, the SenseGlove can reflect force feedback on finger tips of the user. To simulate natural haptic feedback, as an example, at each hand we distribute the estimated force to the 5 fingers by (unequally) dividing the force into 5 separate pieces. The relevant calculation process is shown in Figure 2 (c) and its caption.

#### 4 EXPERIMENT DESIGN AND RESULTS

We employ the “iniVation DAVIS346” 2D event camera for both the frames and events capture (automatically synchronized). We use the SenseGlove Development Kit for haptic feedback, which can simulate vibrotactile and resisting force up to 40 Newtons for each finger. In Figure 3, the first row of pictures shows some captured frames in our testing video scenario; the second row shows the captured (corresponding) events; and the third row shows that Video2Force allows the user to interact with the in-video object by experiencing the dynamic force reflected by SenseGlove, enabling a novel interaction means between the user and the video objects (or video characters).

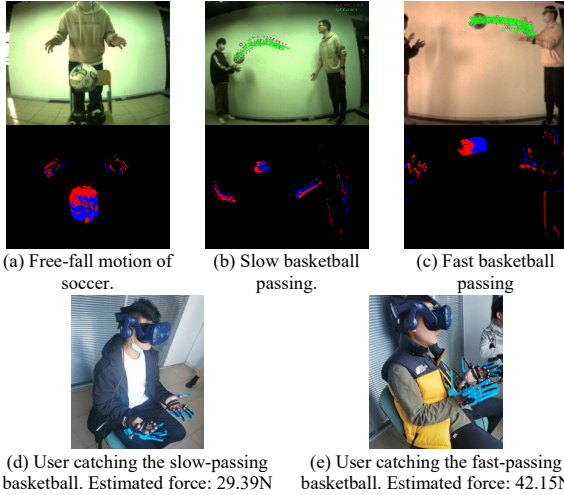


Figure 3: Sample video scenarios: First row shows the captured frames; Second row shows the captured (corresponding) events; Third row shows that the users perceive various magnitudes of forces (reflected by SenseGlove) according to the different speeds of the basketball passing, initiated by the in-video character.

To evaluate the users’ presence with Video2Force, our designed video scenarios are presented to the users with the HTC VIVE Pro Eye head-mounted displays in the following 3 cases: 1) Merely the

Table 1: ANOVA for IPQ Results.

|                     | Observations (Mean±SD) |                     |                      | F      | p       |
|---------------------|------------------------|---------------------|----------------------|--------|---------|
|                     | No Force (n=48)        | Static Force (n=48) | Dynamic Force (n=48) |        |         |
| Spatial Presence    | 3.84±1.29              | 4.83±0.82           | 5.09±0.99            | 18.772 | 0.000** |
| Involvement         | 5.34±1.18              | 5.47±1.06           | 5.38±1.11            | 0.1810 | 0.834   |
| Experienced Realism | 4.08±1.44              | 4.85±0.98           | 5.24±1.11            | 11.643 | 0.000** |
| General Presence    | 4.13±1.66              | 5.02±0.98           | 5.46±1.27            | 12.499 | 0.000** |

\* $p < 0.05$  \*\* $p < 0.01$

video playback without any force feedback (No Force); 2) Video playback synchronized with fixed-level vibrotactile feedback (Static Force); and 3) Video playback synchronized with dynamic force feedback of Video2Force (Dynamic Force). After the video experience, the users are requested to take the igroup presence questionnaire (IPQ) [8]. A total of 48 subjects, recruited from university students and professionals ( $M=23.29$ ,  $SD=1.68$ ), participated into the experiments. It can be observed from the questionnaire results (Table 1) that overall the dynamic force feedback of Video2Force can enhance the user’s presence considerably over the cases of no force and static force. For the factor “involvement”, it is apparent the differences are statistically insignificant between the 3 cases. This may due to that the current SenseGlove is still sort of clumsy, which affects users’ involvement.

#### 5 DISCUSSION AND FUTURE WORK

We present the initial version of Video2Force verified by preliminary experiments. To the best of our knowledge, this is the first work that bridges the estimated video motion and the user’s dynamic force feedback via the emerging event-based processing and novel haptic gloves. In our future work, we plan to improve Video2Force by 1) enhancing event-based tracking with deep learning-based methods; 2) attempting more types of haptic devices to make Video2Force more comprehensive; and 3) augmenting Video2Force for 6 degree-of-freedom light field videos, etc.

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